# MATH441/MATH541
# Applied Mathematics II: Complex Variables
# Lecture slides 5: Exponential and related functions

Nick Strigul
Washington State University Vancouver
nick.strigul@wsu.edu

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# Lecture outline

1. Harmonic functions
2. Exponential functions
3. Trigonometric functions
4. The logarithm
5. The function $z^\alpha$

# Harmonic Functions. Part 1

If we assume the existence and continuity in $D$ of the second order partials derivatives of $u$ and $v$ with respect to $x$ and $y$ (which we will see later is automatically true), then for the Cauchy-Riemann Equations we have

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y}, \qquad \frac{\partial^2 v}{\partial y \partial x} = -\frac{\partial^2 u}{\partial y^2}$$

Then

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

and the same consideration works for $v$:

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Thus $u$ and $v$ both satisfy Laplace's equation (or the potential equation) in two dimensions of the form

$$\nabla^2 \phi = \phi_{xx} + \phi_{yy} = 0$$

# Harmonic Functions. Part 2

## Definition

A function that has second order partial derivatives in a domain $D$ and satisfies Laplace's equation in $D$ is called a *harmonic (or potential)* function.

## Theorem

*Let $u(x, y)$ be harmonic in $D$ and let $N(z_0)$ be any neighborhood of $z_0 = x_0 + iy_0$ that is contained in $D$. Then $u(x, y)$ is the real part of an analytic function that is regular in $N(z_0)$.*

Remark: The three dimensional version of Laplace's equation is

$$\phi_{xx} + \phi_{yy} + \phi_{zz} = 0$$

A solution of this three dimensional equation is called a *logarithmic (or Newtonian) potential.*

# The Exponential Function. Part 1

We desire to extend the real functions such as $e^x$, $\cos x$, $\sin x$ to complex functions. Rather than simply define these complex functions, we take an alternate approach.

Consider $e^x$. We seek a function $f(z)$ such that $f(x) = e^x$ when $x$ replaces $z$. Recall that $f(x) = e^x$ satisfies the initial value problem

$$f'(x) = f(x) \qquad f(0) = 1$$

In an analogous fashion we seek a function $f(z) = e^z$ which is analytic and satisfies

$$\left.\begin{aligned} f'(z) &= f(z) \\ f(0) &= 1 \end{aligned}\right\} \qquad (1)$$

# The Exponential Function. Part 2

If such a function exists, then it will reduce to $e^x$ when we replace $z$ by $x$. Since $f(z) = u(x,y) + iv(x,y)$ then (1) $\Longrightarrow$

$$u_x(x,y) + iv_x(x,y) = u(x,y) + iv(x,y)$$

and therefore that $u_x(x,y) = u(x,y)$ and $v_x(x,y) = v(x,y)$. Solutions to these equations are

$$\left.\begin{aligned} u(x,y) &= p(y)\,e^x \\ v(x,y) &= q(y)\,e^x \end{aligned}\right\} \qquad (2)$$

From $f(0) = u(0,0) + iv(0,0) = 1$, we have that $u(0,0) = 1$ and $v(0,0) = 0$. Hence $p(0) = 1$ and $q(0) = 0$. From the Cauchy-Riemann equations we have that
$u_x(x,y) = p(y)\,e^x = v_y(x,y) = q'(y)\,e^x$ and
$u_y(x,y) = p'(y)\,e^x = -v_x(x,y) = -q(y)\,e^x$

# The Exponential Function. Part 2

Hence $p(y) = q'(y)$ and $q(y) = -p'(y)$ so that

$$p(y) = -p''(y) \qquad \text{and } q(y) = q''(y)$$

Thus both $p$ and $q$ satisfy the harmonic oscillator equation of the form $\frac{d^2\phi}{dy^2} + \phi = 0$. We also have that $p(0) = q'(0) = 1$ and $q(0) = -p'(0) = 0$. The general solution of the harmonic oscillator equation is $A\cos y + B\sin y$. Hence

$$p(y) = \cos y \text{ and } q(y) = \sin y.$$

Therefore

$$f(z) = e^z = e^{x+iy} = e^x(\cos y + i\sin y). \qquad (3)$$

# The Exponential Function. Part 3

## Definition

A function which is regular at all points of the $z-$plane is called an *entire function*. We subdivide entire functions into entire rational functions (only polynomials and their quotients) and entire transcendental functions (all others).

Note: $e^z$ is an entire transcendental function.
Note that if we set $x = 0$ and $y = \theta$ in (3) then

$$e^{i\theta} = \cos\theta + i\sin\theta$$

so that we have a new representation for $z$, namely

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}$$

# The Exponential Function. Part 4

## Theorem

*For the complex exponential we have*

$(a)$ $\quad e^{z_1} e^{z_2} = e^{z_1+z_2}$

$(b)$ $\quad |e^z| = e^x$

$(c)$ $\quad e^{z+2\pi n i} = e^z \; (n = 0, \pm 1, \pm 2, \ldots)$

$(d)$ $\quad e^{2n\pi i} = 1.$

Remark: $e^z = e^{x+iy} = \cos y + i \sin y$ so that $|e^z| = e^x$.
Thus we see that the absolute value $e^z$ is constant on vertical lines. Furthermore, $\arg e^z = y \pmod{2\pi}$, that is the argument of $e^z$ is constant on horizontal lines. Thus we see that $e^z$ is real positive if $y$ is an even multiple of $\pi$, real negative if $y$ is an odd multiple of $\pi$ and pure imaginary if $y$ is an odd multiple of $\pi/2$.

# The Trigonometric Functions. Part 1

In the expression $e^{i\theta} = \cos\theta + i\sin\theta$ first replace $\theta$ by $t$ and then by $-t$ so that we have

$$e^{it} = \cos t + i\sin t \qquad e^{-it} = \cos t - i\sin t$$

Solving for $\cos t$ and $\sin t$ and replacing $t$ by $z$ we define $\cos z$ and $\sin z$ by

$$\left.\begin{aligned} \cos z &= \frac{e^{iz}+e^{-iz}}{2} \\ \sin z &= \frac{e^{iz}-e^{-iz}}{2i} \end{aligned}\right\}$$

$\cos z$ and $\sin z$ are also entire transcendental functions.

# Properties of the Trigonometric Functions. Part 1

Some Properties of $\sin z$ and $\cos z$.

- $\sin z = \sin x \cosh y + i \cos x \sinh y$ and
  $\cos z = \cos x \cosh y - i \sin x \sinh y$

**Proof.**

$$\begin{aligned}
\sin z &= \frac{e^{iz} - e^{-iz}}{2i} \\
&= \frac{1}{2i}\left(e^{ix-y} - e^{-ix+y}\right) \\
&= \frac{1}{2i}\left(e^{-y}\left[\cos x + i \sin x\right] - e^{y}\left[\cos x - i \sin x\right]\right) \\
&= \frac{1}{2i} i \sin x \left(e^{y} + e^{-y}\right) + \frac{1}{2i} \cos x \left(-e^{y} + e^{-y}\right) \\
&= \sin x \cosh y + i \cos x \sinh y
\end{aligned}$$

since $1/i = -i$.

□

# Properties of the Trigonometric Functions. Part 2

Some Properties of $\sin z$ and $\cos z$.:

- $\sin(z + 2n\pi) = \sin z$ and $\cos(z + 2n\pi) = \cos z$
- $\sin z = 0 \iff z$ is an integer multiple of $\pi$.
- $\cos z = 0 \iff z$ is an odd integer multiple of $\pi/2$.
- $\sin(z + w) = \sin z \cos w + \cos z \sin w$
  $\cos(z + w) = \cos z \cos w - \sin z \sin w$
- $\cos(-z) = \cos z$ and $\sin(-z) = -\sin z$
- $\sin^2 z + \cos^2 z = 1$
- $\frac{d}{dz} \sin z = \cos z$ and $\frac{d}{dz} \cos z = -\sin z$.

# The Logarithm. Part 1

Remark: In real calculus the natural logarithm is the inverse of the exponential function. That is, for a positive number $x$,

$$y = \ln x \qquad \Longleftrightarrow \qquad x = e^y.$$

We can think of the transcendental real natural logarithm as a solution of the equation $x = e^y$ for $y$ in terms of $x$.
We may use this approach in seeking a definition of the complex logarithm.
Suppose that $z \neq 0, \infty$. We want to solve for all values of $w$ (if any), such that

$$e^w = z.$$

# The Logarithm. Part 2

Writing $z$ in polar form, namely, $z = re^{i\theta}$ where $r = |z|$ and $\theta$ is any argument of $z$, letting $w = u + iv$, leads to

$$e^{u+iv} = e^u e^{iv} = re^{i\theta}$$

Since $\theta$ and $v$ are real, $\left|e^{i\theta}\right| = \left|e^{iv}\right| = 1$, so that taking the magnitude of the equation above leads to $r = e^u$ or $u = \ln r$, which is the real natural logarithm of the positive real number $r$. Also, $e^{iv} = e^{i\theta}$, so that $v = \theta + 2n\pi$, where $n$ is any integer.

In summary, given any complex number $z \neq 0, \infty$ there are infinitely many numbers $w$ such that $e^w = z$; all such numbers are contained in the expression

$$\ln|z| + i\theta + 2n\pi i = \ln|z| + i \arg z$$

# The Logarithm. Part 3

## Definition

Each of the solutions $w$ of $e^w = z \quad (z \neq 0, \infty)$ is called a *logarithm of* $z$. The function which associates with each such $z$ the corresponding values of $w$ is called the logarithm of $z$ and is denoted by $w = \log z$.

Hence

$$\log z = \ln |z| + i \arg z \qquad (z \neq 0, \infty)$$

where $\ln |z|$ denotes the real logarithm of $|z|$ and $\arg z$ is given all admissible values. since values of $\arg z$ differ by multiples of $2\pi$, it follows that the various determinations of $\log z$ differ by multiples of $2\pi i$.

## Definition

The *principal value* or *determination* of the logarithm corresponds to the principal determination of the argument. We shall define it by the restriction $-\pi < Im(\log z) = \theta \leq \pi$. It is often denoted by $Log\, z$.

# The Logarithm. Part 4

## Example

Consider $z = 1 + i$. Then $|z| = \sqrt{2}$ and $\arg z$ consists of all numbers $\pi/4 + 2n\pi$, where $n$ is any integer. Hence

$$\log(1+i) = \ln\sqrt{2} + i\left(\tfrac{\pi}{4} + 2n\pi\right)$$

The principal value of $\log(1+i)$ is $Log\,(1+i) = \ln\sqrt{2} + i\frac{\pi}{4}$.

Remark: $\log(z_1 z_2) - \log z_1 - \log z_2 = 0 \qquad (mod\, 2\pi i)$

Remark: One may verify by direct computation that $\log z = \log\sqrt{x^2+y^2} + i\theta$ satisfies the logarithm Cauchy-Riemann equations at all finite $z \neq 0$. Also, one can show that

$$\frac{d}{dz}(\log z) = \frac{1}{z}$$

# The Logarithm. Part 5

Additional comments: In each determination $\log z = \ln|z| + i\theta$, where the *nth* determination is defined by $(2n-1)\pi < \theta \leq (2n+1)\pi \quad (n = 0, \pm 1, \pm 2, \ldots)$. The value $\theta = 2n\pi$ for $z$ real and positive reduces $\log z$ to $\ln x$, the real natural logarithm. The extension to an analytic function is unique in that the various values of $\log z$ must be regarded as different determinations of a *single analytic function* which, however, is multiple valued and has a *singularity at the origin.* ($z = 0$ is called a branch point)

Indeed, the extension of the real function $\ln x$ (starting, say, from $\theta = 0$) yields any and all of the determinations of $\log z$ by so-called "analytic continuation" (by means of overlapping neighborhoods) around the origin.

The principal determination corresponds to a "cutting" of the plane along the negative real axis. This "cutting" of the plane restricts $\log z$ to a *single-valued* analytic function (once the value is prescribed at any point $z \neq 0$.) The curve along which this cur may be made is not unique. Any non-self-intersecting curve from $z = 0$ to $\infty$ will do, provided exactly one "edge" of the cut is considered as belonging to the cut plane.

# The Function $z^\alpha$. Part 1.

If $\alpha$ is an arbitrary constant, then $z^\alpha$ is defined by

$$z^\alpha = e^{\alpha \log z} \qquad (z \neq 0, \infty)$$

In general $z^\alpha$ is a multivalued.
Recall that the principal branch (or value) of $\log z$ is denoted by $Log\, z$ where $Log\, z = \ln|z| + i\theta \quad (-\pi < \theta \leq \pi)$. The *nth* branch of $\log z$ is $\log z = Log\, z + 2n\pi i \qquad (n = 0, \pm 1, \pm 2, \ldots)$. Hence

$$(z^\alpha)_n = e^{\alpha Log\, z} e^{2\pi n i}$$

## Theorem

*$z^\alpha$ is an $n-$valued function ($n$ a positive integer)$\Longleftrightarrow$ $\alpha$ is a (real) rational number of the form $\frac{m}{n}$, where $m$ and $n$ have no common factor.*

## Example

$z^{\frac{3}{10}}$ is a $10-$valued function.

# The Function $z^\alpha$. Part 2.

## Proof.

If $\alpha = \frac{m}{n}$, then for any $z \neq 0$ the numbers $e^{2\pi i(\frac{m}{n})k}$ $(k = 0, \pm 1, \pm 2, \ldots)$ assume exactly $n$ distinct values; e.g. for $k = 0, 1, \ldots n-1$. For if $k' = k + jn$, where $j$ is an integer, then

$$e^{2\pi i(\frac{m}{n})k\prime} = e^{2\pi i(\frac{m}{n})k} e^{2\pi i(\frac{m}{n})jn} = e^{2\pi i(\frac{m}{n})k}$$

since $e^{2\pi imj} = 1$ since $jm$ is an integer.

Converse: If $z^\alpha$ is $n-$valued, then $\exists$ at least two distinct $k's$ say, $k_1$ and $k_2$ such that

$e^{2\pi i\alpha k_1} = e^{2\pi i\alpha k_2} \implies e^{2\pi i\alpha(k_1-k_2)} = 1 \implies \alpha(k_1 - k_2) = l$ where $l$ is an integer, since $e^{2\pi ir} = 1 \iff r$ is an integer. Thus $\alpha = \frac{l}{k_1-k_2} = \frac{m}{p}$, where $m$ an $p$ have no common factors.

But $p = n$; for if $p < n$ then the first part of the proof implies that $z^\alpha$ is $p$ valued contradicting the assumption of $n$ valuedness. If $p > n$ then again $z^\alpha$ is $p$ valued, contradicting $n$ valuedness. Hence $p = n$.

$\square$

# The Function $z^\alpha$. Part 3

**Theorem**

*If $\alpha$ is not rational, $(z^\alpha)_k$ $(k = 0, \pm 1, \pm 2, \ldots)$ assumes a different value for each $k$.*

**Proof.**

If $\exists$ distinct $k_1$ and $k_2$ such that $e^{2\pi i \alpha k_1} = e^{2\pi i \alpha k_2}$, then we conclude as before that $\alpha$ is rational, which is a contradiction. $\square$

Remark: If $\alpha$ is not rational, we have two cases:
a) $\alpha$ is real and irrational. Then $\left|e^{2\pi i \alpha k}\right| = 1$ for all $k$ and therefore $|z^\alpha| = \left|e^{\alpha(\ln|z| + 2\pi i k)}\right| = e^{\alpha \ln|z|} = |z|^\alpha$
b) If $\alpha$ is not real, then $\alpha = \alpha_r + i\alpha_i$ so that $\left|e^{2\pi i(\alpha_r + i\alpha_i)k}\right| = e^{-2\pi\alpha_i k}$ which is different for each value of $k$.

# The Function $z^\alpha$. Part 4.

Additional comments on using certain familiar results.
1) $z^\alpha z^\beta = e^{\alpha \log z} e^{\beta \log z}$ and $e^{\alpha+\beta} = e^{(\alpha+\beta)\log z}$
These two expressions are equal if $(\alpha+\beta)\log z = \alpha \log z + \beta \log z$
which holds if proper determinations of $\log z$ are used. It is certainly true if the same determination is used throughout. Other possibilities

$$\alpha(Log z + 2\pi l i) + \beta(Log z + 2\pi m i) = (\alpha+\beta)(Log z + 2\pi n i)$$

which is true if

$$\alpha \cdot 2\pi l i + \beta \cdot 2\pi m i = (\alpha+\beta) \cdot 2\pi n i \text{ or } \alpha l + \beta m = (\alpha+\beta) n$$

Thus $z^\alpha z^\beta = z^{\alpha+\beta}$ provided the above considerations hold.

# The Function $z^\alpha$. Part 5.

Additional comments on using certain familiar results.
2) $(z^\alpha)^\beta = z^{\alpha\beta} = \left(z^\beta\right)^\alpha$

## Example

Example of 2) If $x$ is real and positive, then $\left(x^{\frac{1}{2}}\right)^3 = \left(x^3\right)^{\frac{1}{2}} \Longleftrightarrow$the proper roots (proper branches of the power function) are chosen. Let $x = 1$, then $x^{\frac{1}{2}}$ may be $-1$ so that $\left(x^{\frac{1}{2}}\right)^3 = -1$, whereas $x^3 = 1$ and $\left(x^3\right)^{\frac{1}{2}} = (1)^{\frac{1}{2}}$ may be $+1$.

## Example

Find the principal value of $(-i)^i$.

$(-i)^i = e^{iLog(-i)} = e^{iLog\left(e^{-\frac{\pi i}{2}}\right)} = e^{i\left(-\frac{\pi i}{2}\right)} = e^{\frac{\pi}{2}}$

Remark: $\frac{d}{dz}\left(z^\alpha\right) = \frac{d}{dz}\left(e^{\alpha \log z}\right) = e^{\alpha \log z}\frac{\alpha}{z} = \alpha z^\alpha z^{-1} = \alpha z^{\alpha-1}$
provided the proper determinations of log are used.

# Thank you